

Term 7- Sept 2025

Nanoelectronics and Technology
(01.119/99.503)-Week 4 Class 1

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07-Oct- 2025

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Outline

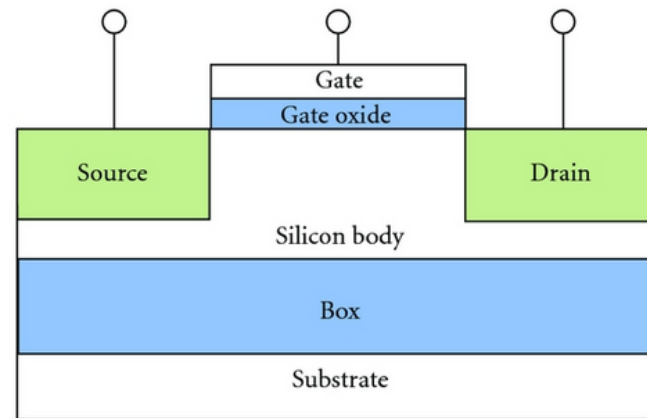
- CMOS Scaling Challenges
- FinFETS
- Interconnect Scaling and Issues

CMOS scaling: Challenges

- Gate oxide scaling
 - Leakage current
 - DIBL (Drain Induced Barrier Lowering)
 - GIDL (Gate Induced Drain Leakage)
 - Threshold voltage variation
 - Shallow S/D
 - Mobility issues/Velocity saturation/Hot carrier effect
-
- **Incorporation and integration of new materials**
 - **Thermal budget related issues**
 - **Reliability and Yield related issues**

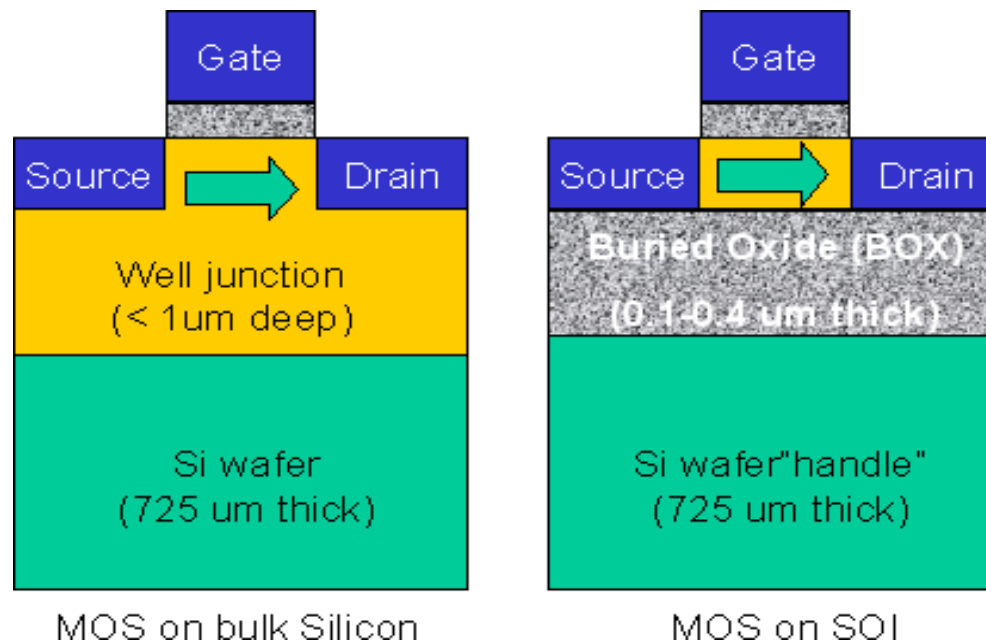
Why SOI

- With Silicon-On-Insulator (SOI) wafers, transistors are formed in thin layers of silicon that are isolated from the main body of the wafer by a layer of electrical insulator, usually silicon dioxide.
- The silicon layer thickness ranges from several microns for electrical power switching devices to less than 50 nm for high-performance microprocessors.
- Isolating the active transistor from the rest of the silicon substrate reduces the electrical current leakage that would otherwise degrade the performance of the transistor.



<https://doi.org/10.1155/2012/565827>

Silicon-On-Insulator (SOI) MOSFET

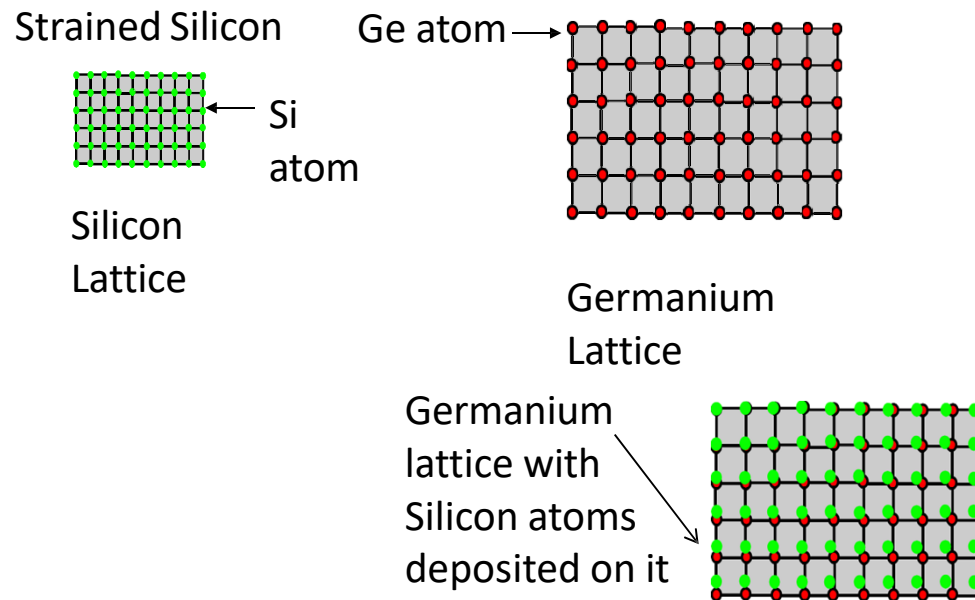


Silicon-on-Insulator (SOI) wafers consist of three layers: a thin (20 nm to several microns, depending on the application) layer of single-crystal silicon on a thick (100 to 400 nm) silicon dioxide layer that is bonded to a conventional "handle" wafer. The entire transistor is located in the thin top layer of silicon and electrically isolated from the bulk wafer by the buried oxide (BOX) layer.

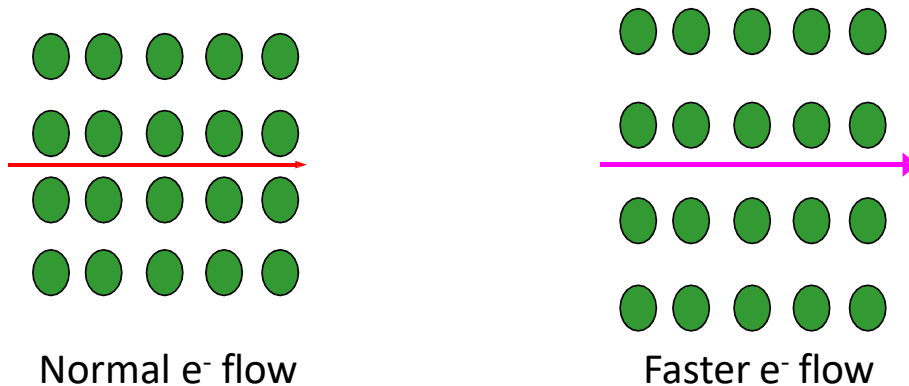
Strained Silicon Wafers

Strained-Si on SiGe virtual substrate:

- Grow a relaxed SiGe layer → lattice mismatch stretches thin Si grown on top.
- Provides **tensile strain** → boosts **NMOS electron mobility**.
- Used in early 90 nm–65 nm processes.



Higher Mobility in Strained Si

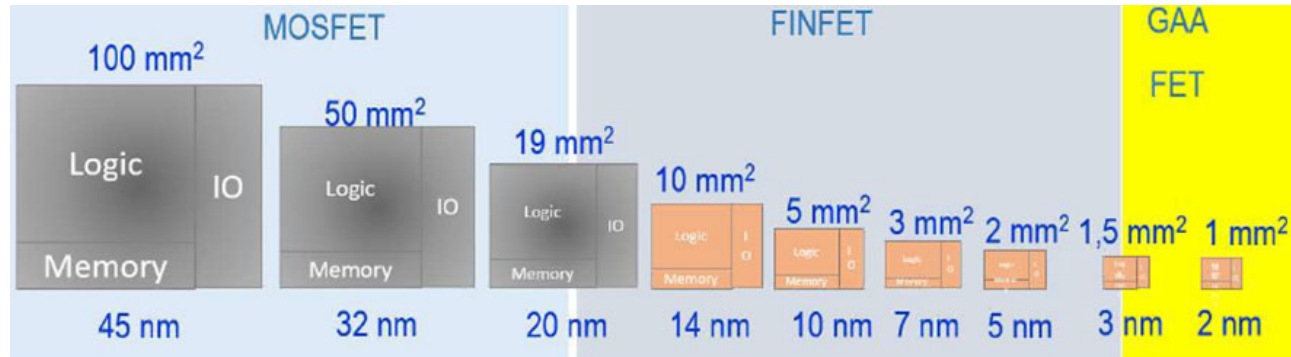


Stressing, stretching, or straining, the Si lattice allows electrons to flow with less resistance. Lower resistance means that transistors switch faster.

FinFETs

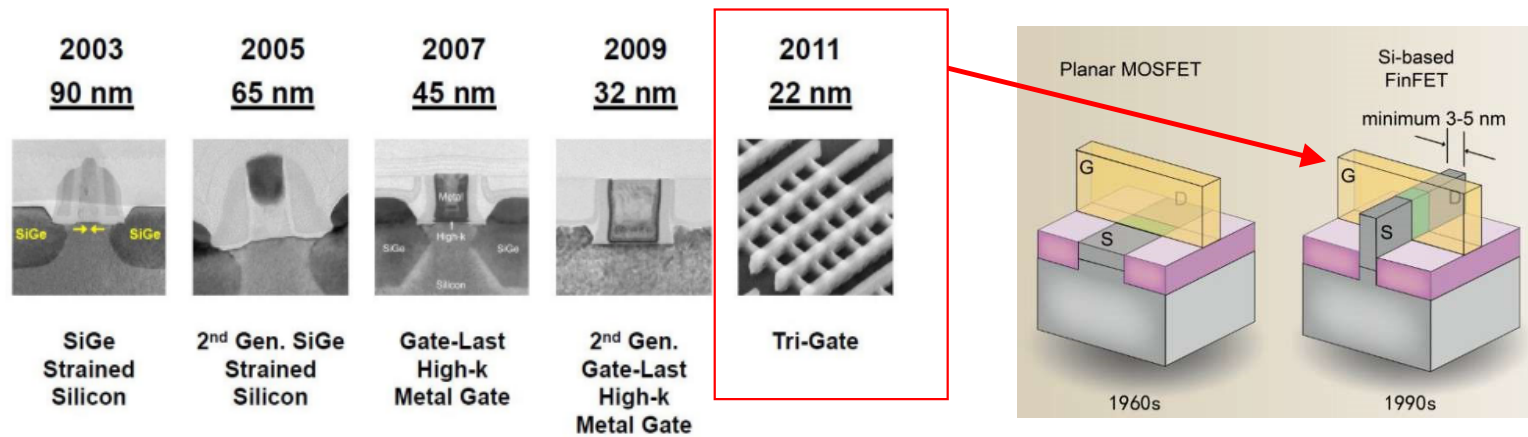
- Known by different names within different companies.
 - Tri gate
 - FinFET
 - Dual gate
- The term FinFET was coined by **Prof. Chenming Hu** **University of California, Berkeley** to describe a nonplanar, double-gate transistor built on an SOI substrate. (1998)
- The distinguishing FinFET characteristic is the elevated conducting channel above the base Si wafer that is wrapped by a thin silicon "fin", forming the device gate.
- Tri-gate or FinFet are terms used by Intel Corporation to describe their non planar transistor architecture for use in **22nm microprocessor technologies**.

FinFETs



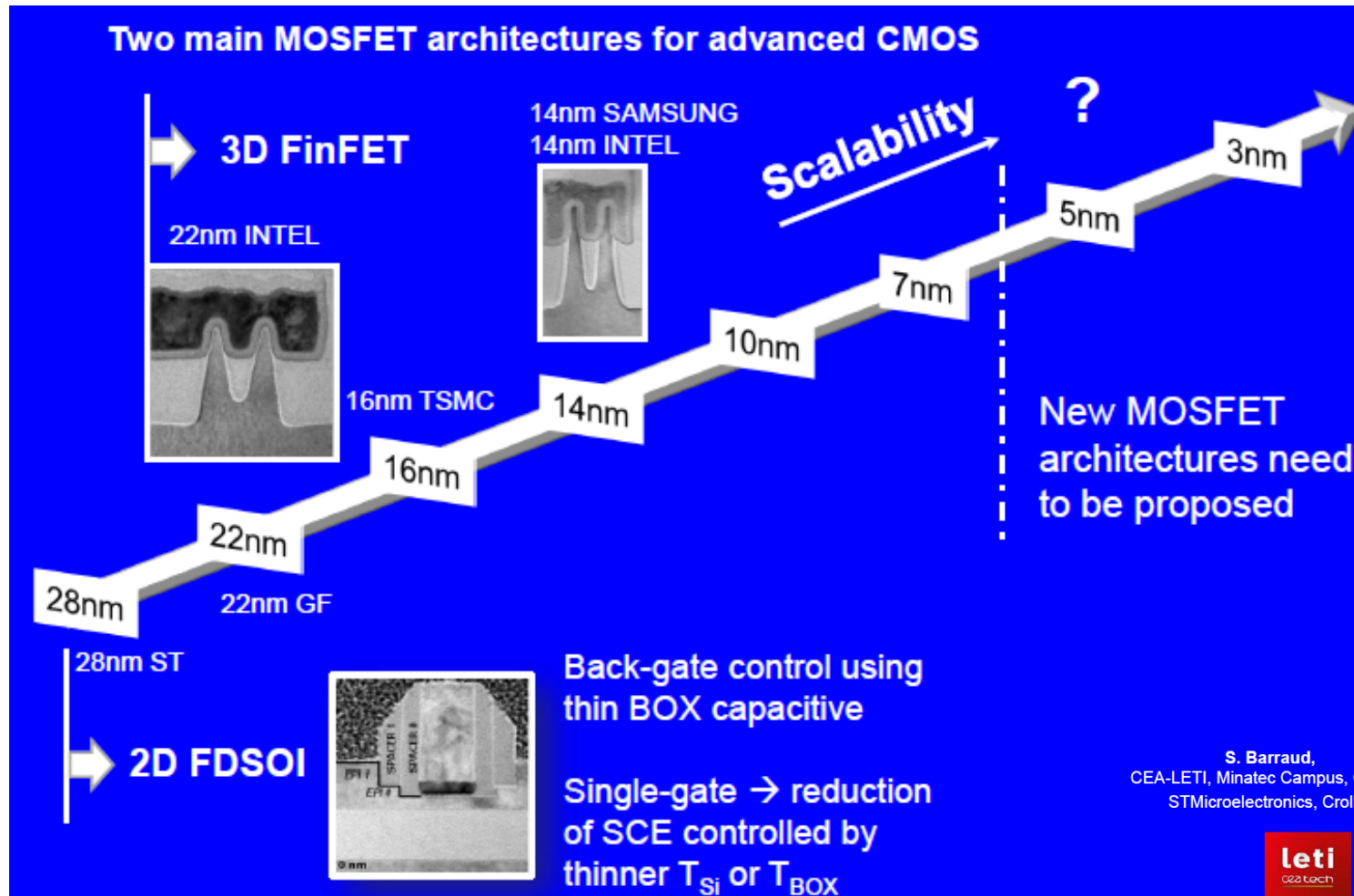
Adapted from Mistry, K. (2017). 10 nm technology leadership, Technology and Manufacturing Day, Intel. 2017.

Scale down benefits in terms of silicon integration, and associated FET technology.



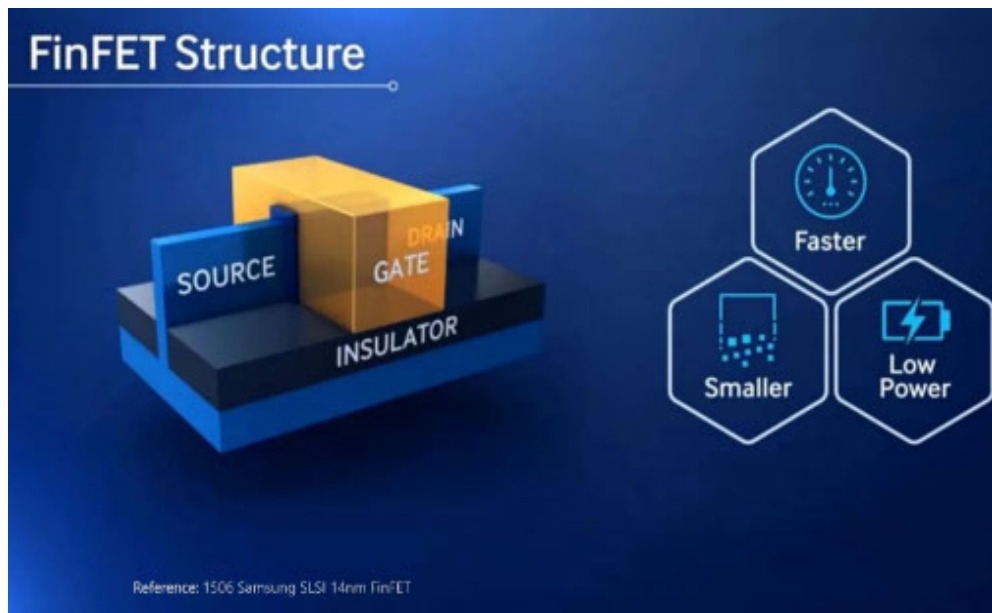
Ref: K. Mistry, "Tri-Gate Transistors: Enabling Moore's Law at 22nm and Beyond", SEMICON West, July 2014.

FinFETs



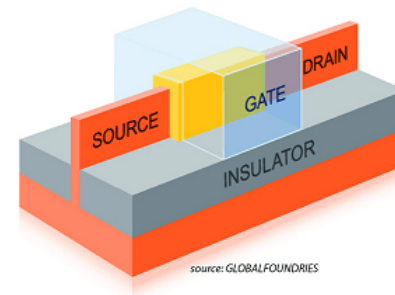
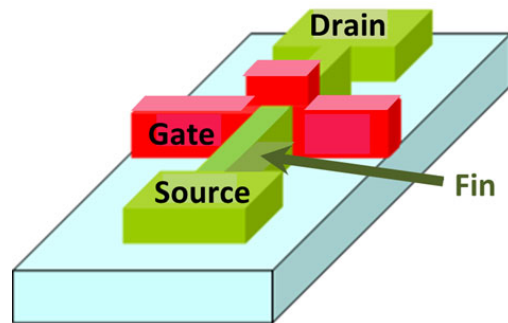
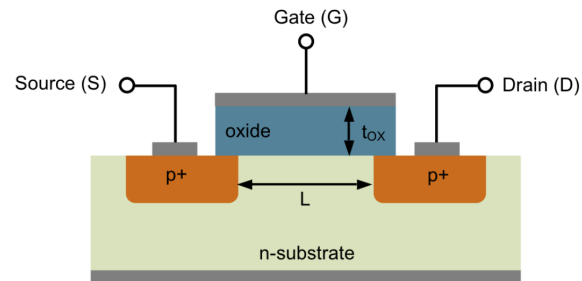
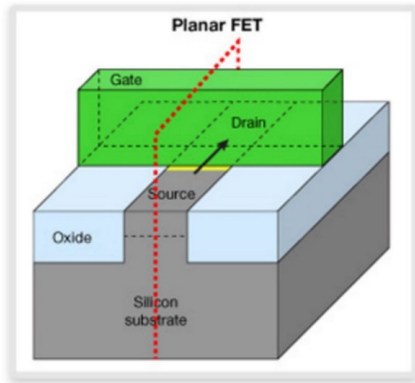
FinFETs

A fin field-effect transistor (FinFET) is a **multigate device**, a MOSFET (metal-oxide-semiconductor field-effect transistor) built on a substrate where the gate is placed on two, three, or four sides of the channel or wrapped around the channel, forming a double or even multi gate structure.

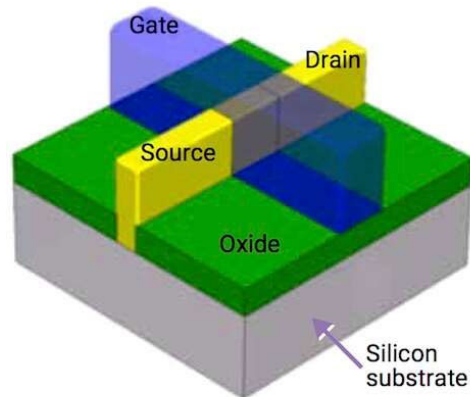


- Significant increase in
- Reduction in power consumption
- Higher drive current
- However, cost increase in process

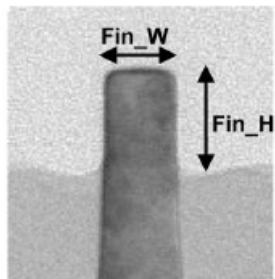
Planar MOSFET and FinFETs



FinFETs



a



$$W_{\text{eff}} = 2 \times \text{Fin_H} + \text{Fin_W}$$

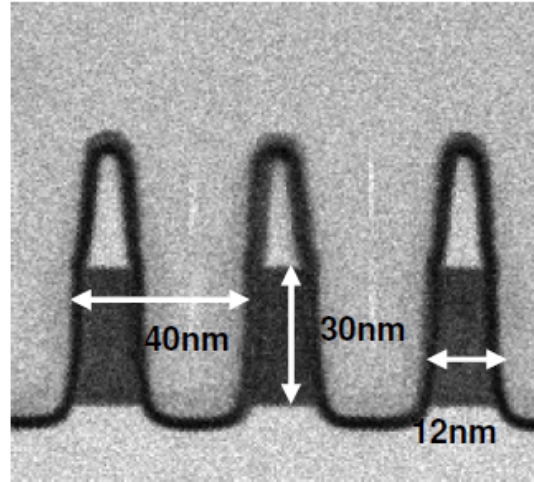
Chenming Hu, August 2011

Structure of FinFETs

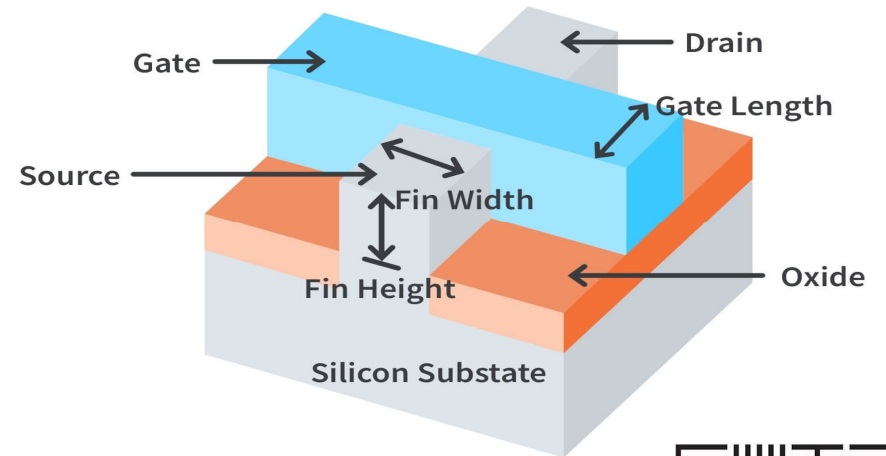
- 1. Fin Design:** The key feature of FinFETs is their "fin" structure, which is a vertical channel that resembles a fin. This three-dimensional geometry increases the surface area for electrostatic control over the channel.
- 2. Gate Configuration:** In FinFETs, the gate surrounds the channel on three sides (top and both sides), providing better electrostatic control compared to traditional planar transistors, which have a gate on only one side.
- 3. Channel Material:** FinFETs can use various semiconductor materials, typically silicon, but advanced designs might use silicon-germanium or other materials to enhance performance.

FinFETs

Vdd (V)	0.9
Lg (nm)	20-30
Gate Pitch (nm)	80-100
Fin Pitch (nm)	40
Dfin (nm)	12
Hfin (nm)	30

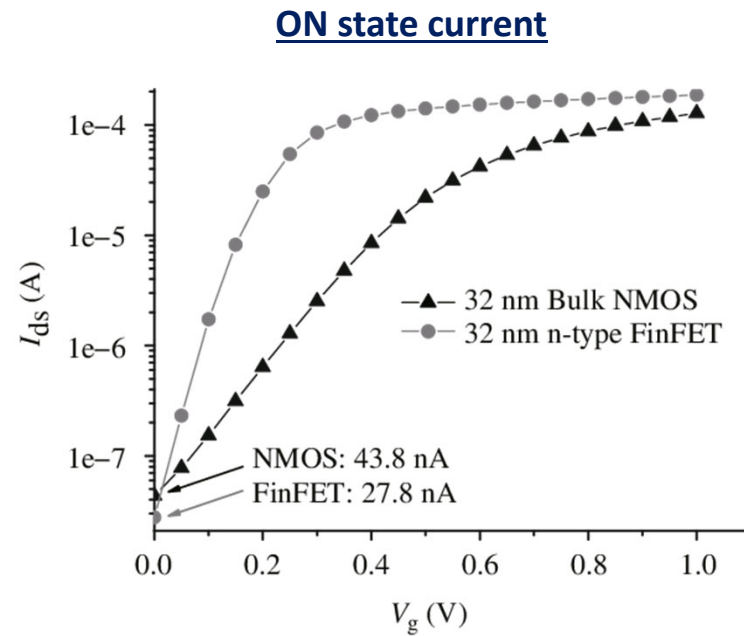
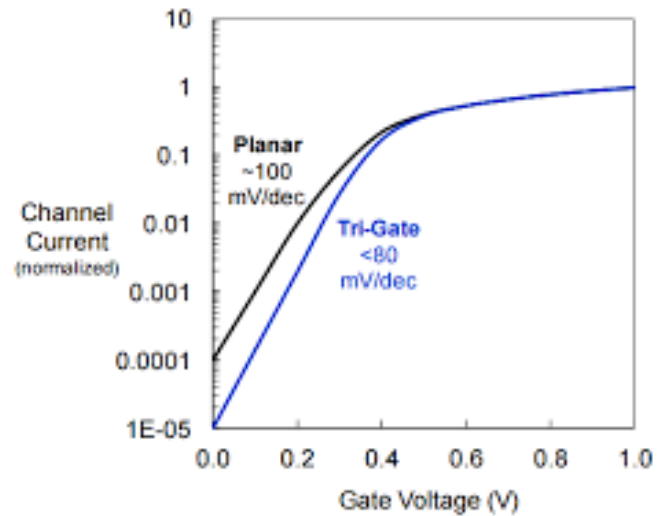


Relevant 22nm SOI-FinFET device parameters used in this study.



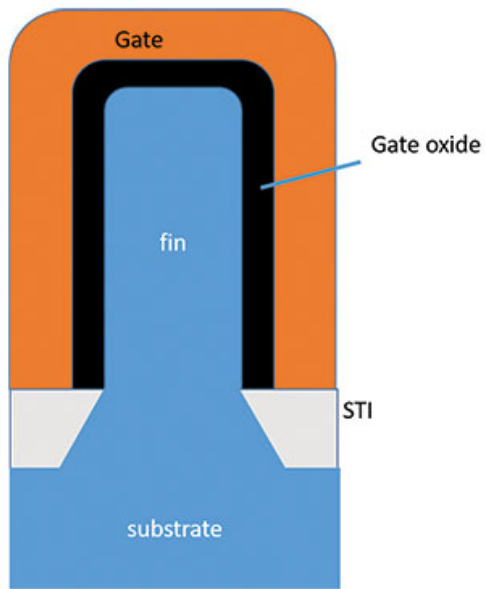
Ref: "Channel Doping Impact on FinFETs for 22nm and Beyond", C. -H. Lin, et al., VLSI 2012

FinFETs

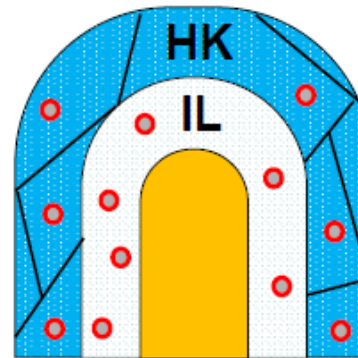


Ref: "Low power adiabatic logic based on FinFETs", L. Nan, 2014, DOI: [10.1007/s11432-013-4902-x](https://doi.org/10.1007/s11432-013-4902-x)

FinFET : Challenges



FinFET Device

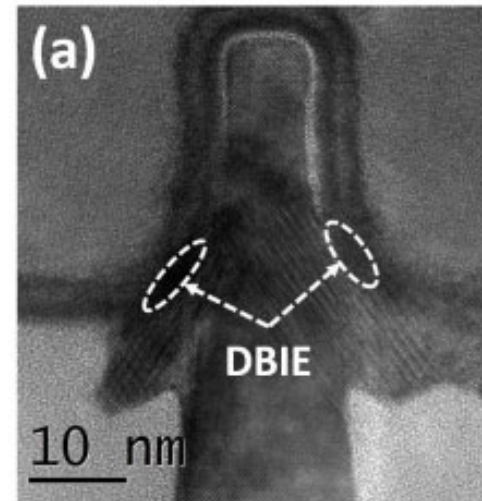


Different defect generation probability

Geometrical Impact (non-uniform field distribution)

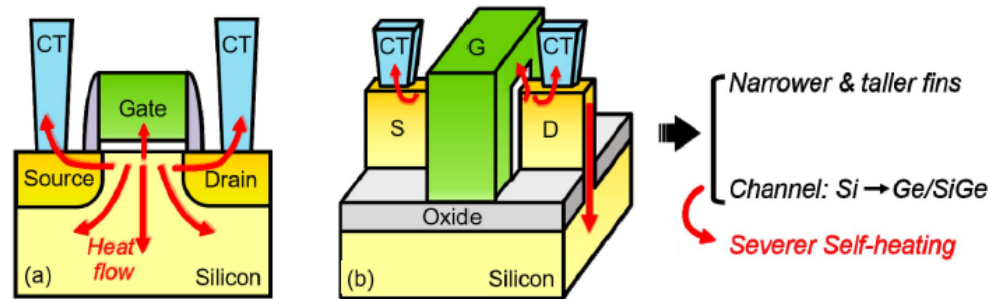
Heat confinement

- Non-uniform electric field distribution
- High electric field at the corner regions
- Self Heating



Ref: Sen Mei, PhD thesis, SUTD (2015/2016) Characterization and Simulation Study of FEOL Reliability in Advanced Logic and Memory Devices

FinFET : Challenges



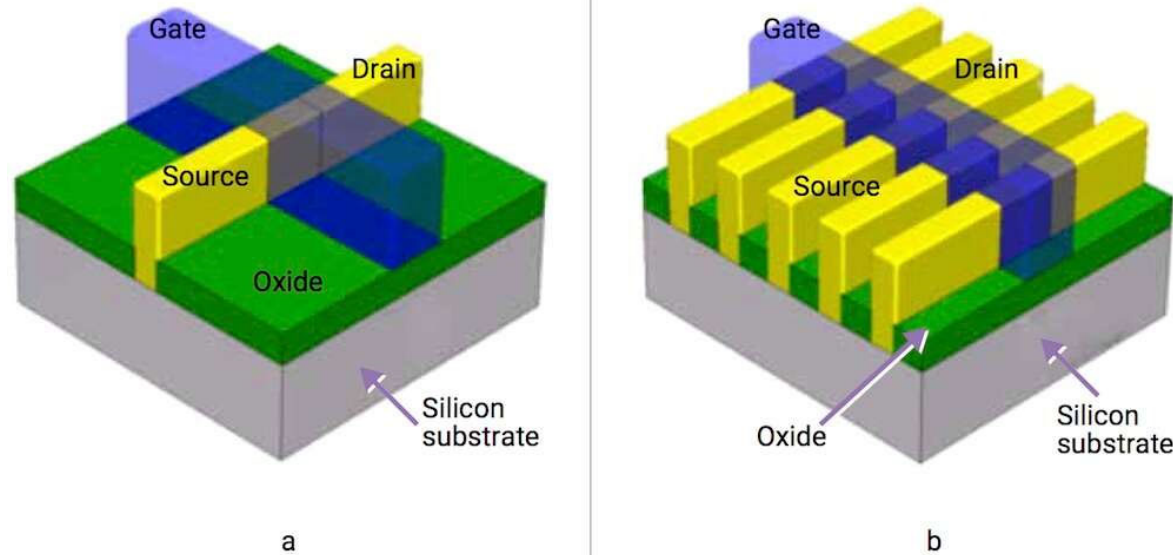
Self Heating:

- Expecting higher temperatures for new FinFET devices
- Shorter gates and higher fins increase temperature
- Higher channel temperature for higher power consumed

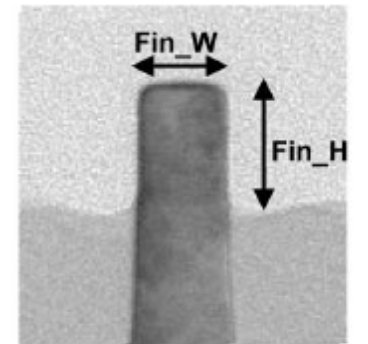
Performance and Reliability challenges:

- Performance degrades due to thermal impact
- Higher temperatures degrades device life time

FinFETs



- A **multi-fin FET** simply places multiple fins in parallel under a single gate.
- Each fin acts as a conduction channel, and the total current is the **sum of all fin currents**.
- To get enough drive strength for modern logic gates, designers replicate fins under one gate.



$$W_{eff} = 2 \times Fin_H + Fin_W$$

Chenming Hu, August 2011

FinFET : Challenges

- Precise etching of Fins and Control of the Fin height
- Fin integration (Collapse and erosion)
- Fabrication complexity due to multi-patterning and increase in manufacturing cost
- Higher parasitic capacitance due to 3D profile of the FinFET devices

FinFET to GAA Transistor Evolution

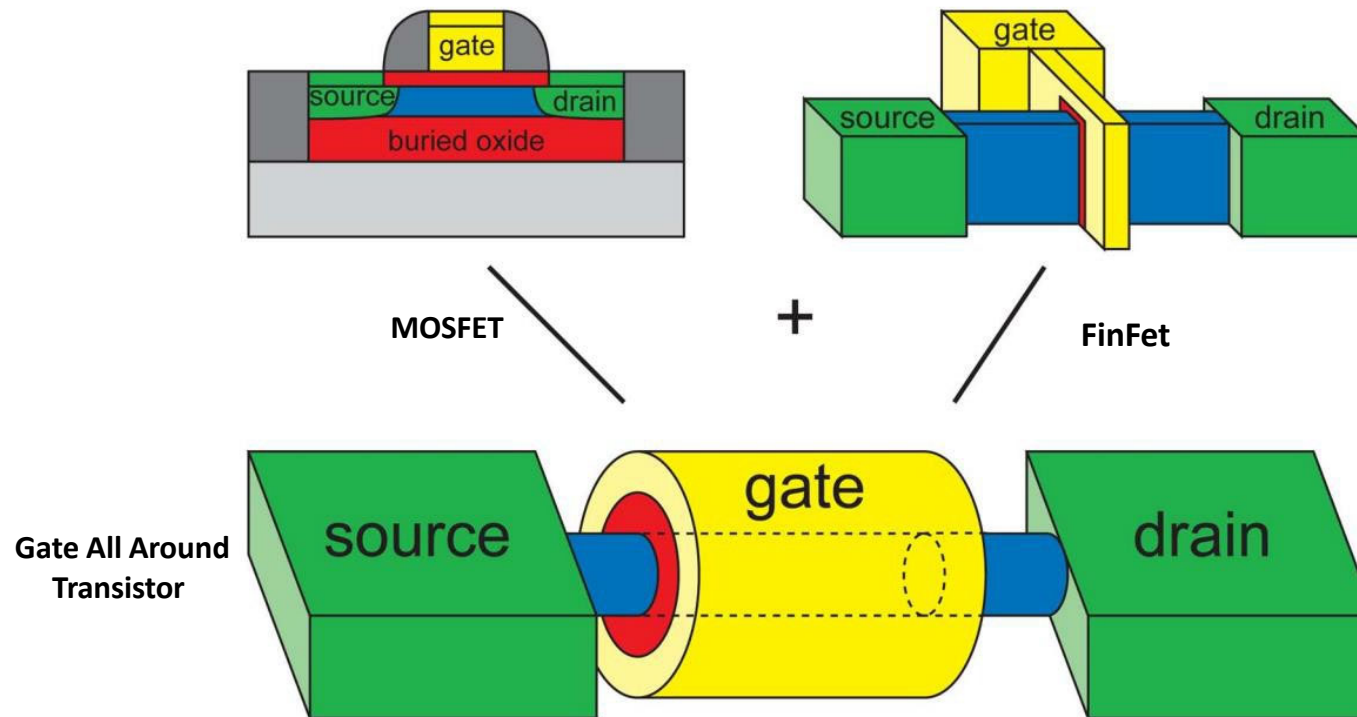


Fig. 3. Combining the right type of high-mobility channel material with an ultrathin body semiconductor allows for improved on-state performance.

Using GAA gives you a 10% increase in performance and a 30% decrease in power consumption compared to FinFets

Nanosheet FET Transistor



- Typically, **3–4 stacked nanosheets per device** to provide enough drive current.
- Similar to lateral nanowire FETs, nanosheet FETs use wider and thicker wires to provide improved electrostatics and drive current.
- Providing tunable channel width (sheet width can be varied for power/performance trade-offs).

Typical Ion/Ioff Ratios by Node

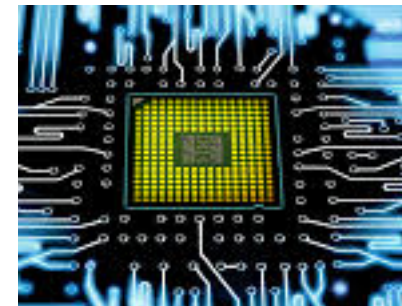
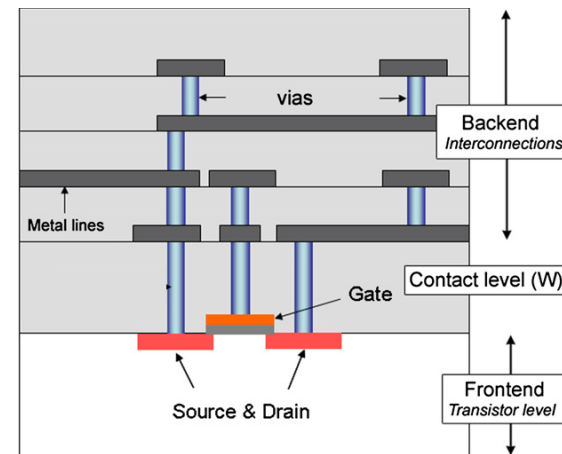
CMOS Node	Era	Ion/Ioff Ratio (Approx.)	Notes
180 nm	~1999	10^6 – 10^7	Long channels, low leakage
130 nm	~2001	10^6	Good balance, leakage manageable
90 nm	~2003	10^5 – 10^6	Leakage starts rising (subthreshold + tunneling)
65 nm	~2006	10^5	Gate leakage significant, high-k/metal gate needed
45 nm	~2008	10^5	Strained Si, HKMG introduced (Intel 2007)

Typical Ion/Ioff Ratios by Node

CMOS Node	Era	Ion/Ioff Ratio (Approx.)	Notes
32/28 nm	~2011	10^4 – 10^5	Short channel effects, leakage trade-offs
22 nm (FinFET)	~2012	10^5 – 10^6	FinFET gives excellent control, better Ion/Ioff
14/16 nm	~2014	10^5 – 10^6	Multi-fin devices, optimized for low leakage
10 nm	~2017	10^5 (logic) / 10^6 (LP)	Leakage again challenging
7 nm	~2019	10^5 – 10^6	FinFET scaling limits, multiple V_T devices
5 nm	~2020	10^5 – 10^6	Advanced FinFET, EUV lithography
3 nm (GAA nanosheet)	~2023+	10^6 – 10^7	Excellent electrostatics, leakage reduction

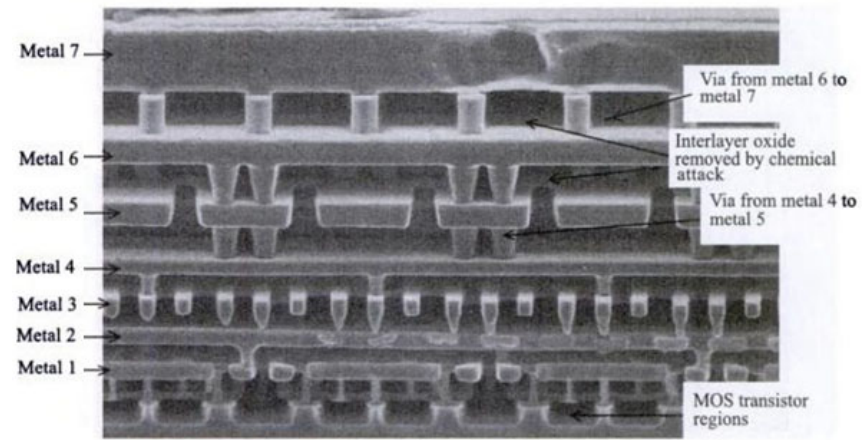
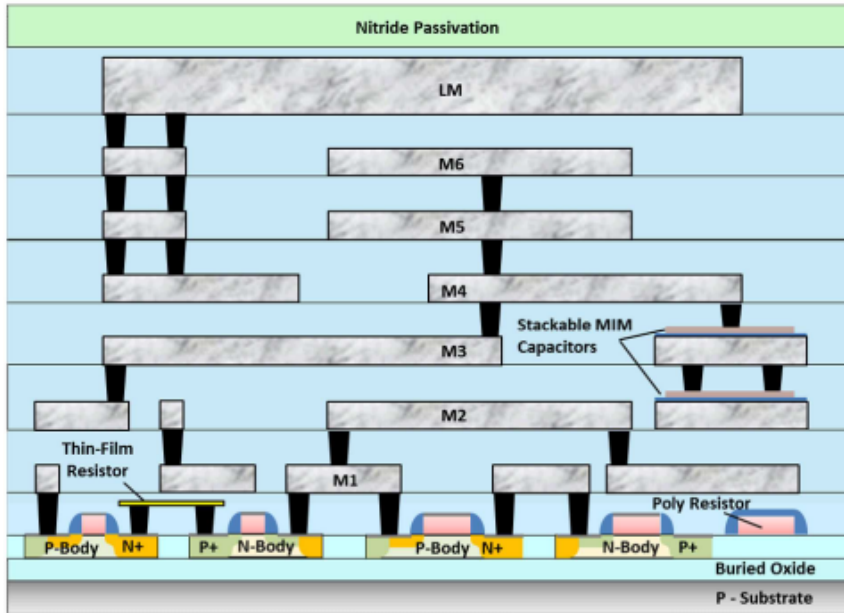
Interconnect Technology

- The goal in low power VLSI design is to combine circuit blocks into an integrated circuit while keeping power consumption and delay times low.
- Interconnects should be designed with optimized power flow to ensure signals transition through an integrated circuit.
- Although there are many solutions to keeping power low, interconnects need to be designed to balance power draw, propagation delay, and SNR.



https://www.researchgate.net/figure/Schematic-cross-section-of-very-large-scale-integrated-VLSI-semiconductor-circuit_fig2_243357711

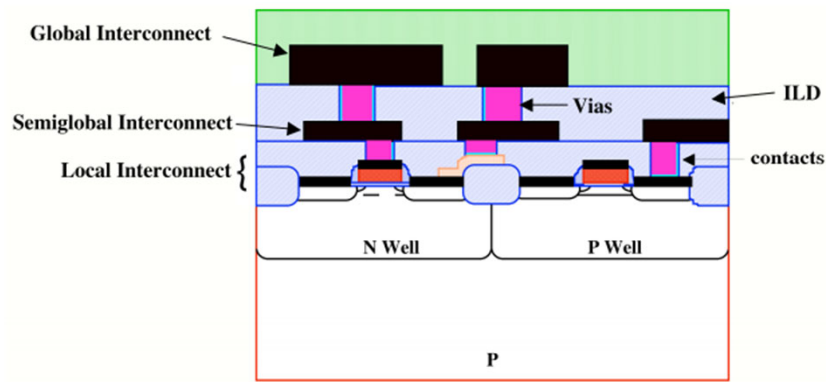
Interconnect Technology



Cross-section of a 0.12 μm technology (Courtesy: Fujitsu)

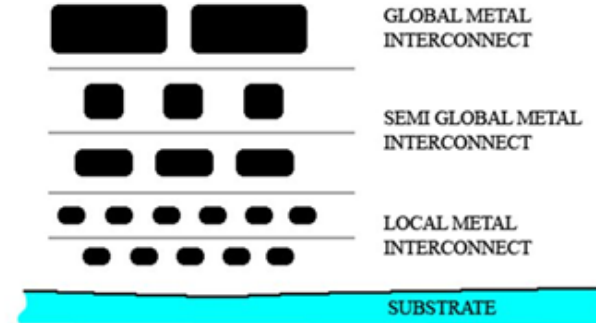
Alan Mitchell, The History of Silicon Scaling: Abridged.

Interconnect Technology

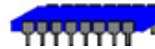


EE 311 / Saraswat Interconnect Scaling

Interconnects in chips are routed in several layers horizontally and vertically and used according to their application

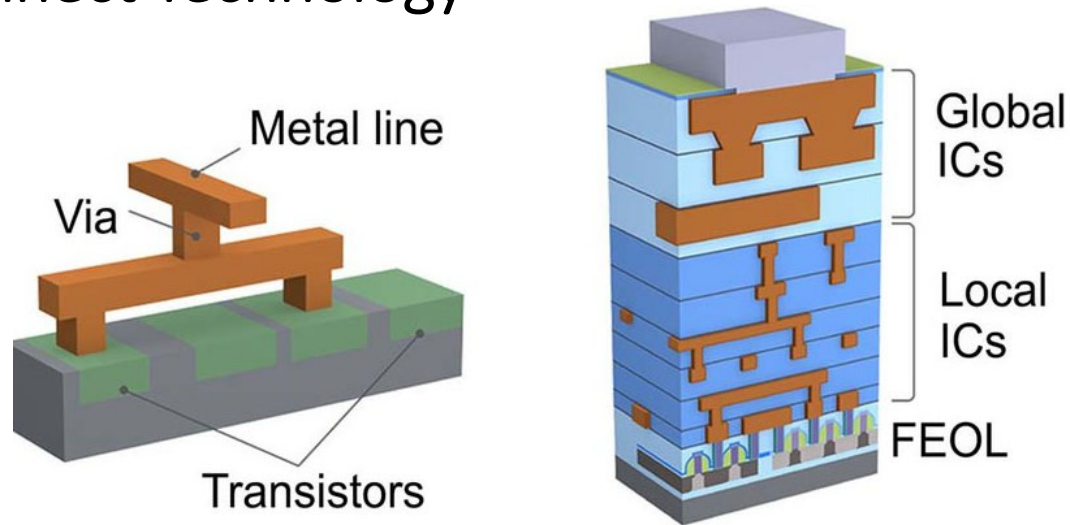


CONCORDIA
VLSI DESIGN LAB



https://www.researchgate.net/figure/Schematic-cross-section-of-very-large-scale-integrated-VLSI-semiconductor-circuit_fig2_243357711

Interconnect Technology



https://www.researchgate.net/figure/General-structure-of-an-IC-with-BEOL-evidenced-a-SEM-section-of-an-Intel-Broadwell_fig1_329551868

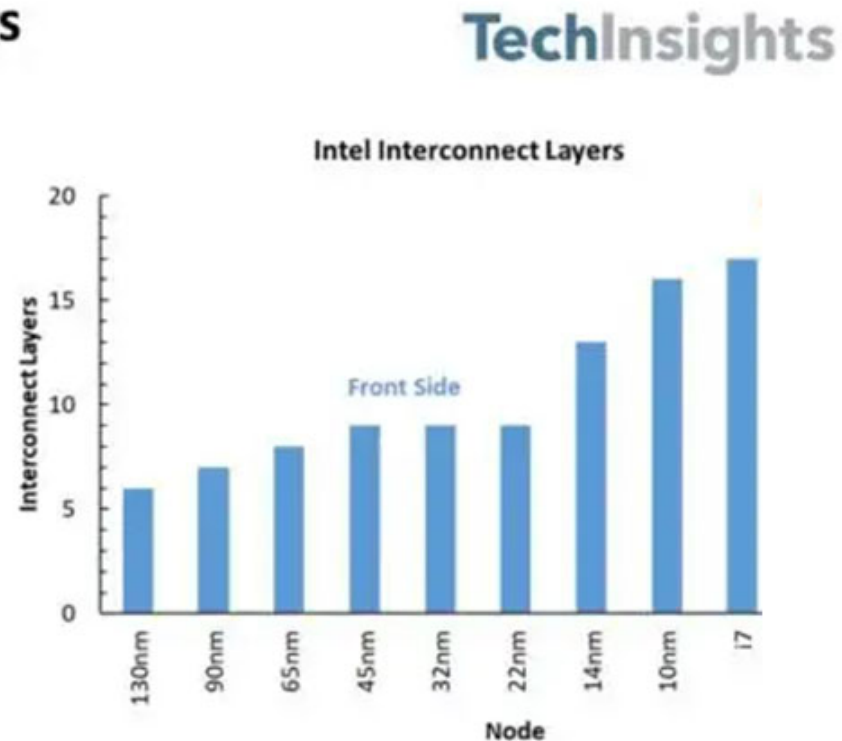
- Local interconnects – used for very short interconnects at the device level
- Semiglobal interconnects – Used to connect devices within a block
- Global interconnects – Used to connect long interconnects between the blocks, including power, ground and clocks.

Ref: Prof. Saraswat, EE 311 Notes on interconnects, Stanford University

Interconnect Technology

Intel Interconnect Layers

- New nodes drive increasing interconnect counts.
- Source TechInsight reverse engineering analysis and Intel papers.



<https://semiwiki.com/semiconductor-manufacturers/330343-vlsi-symposium-intel-power-via/>

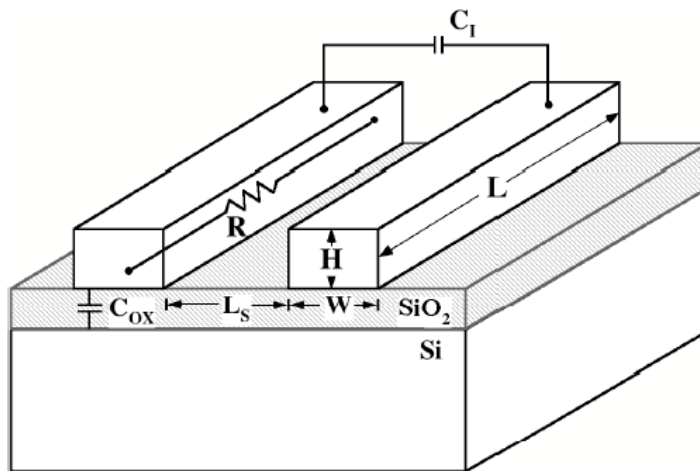
Interconnect Scaling

Two drivers for Interconnect Scaling:

- RC delay
- Electro-migration
- Stress-migration

Interconnect Scaling

RC delay



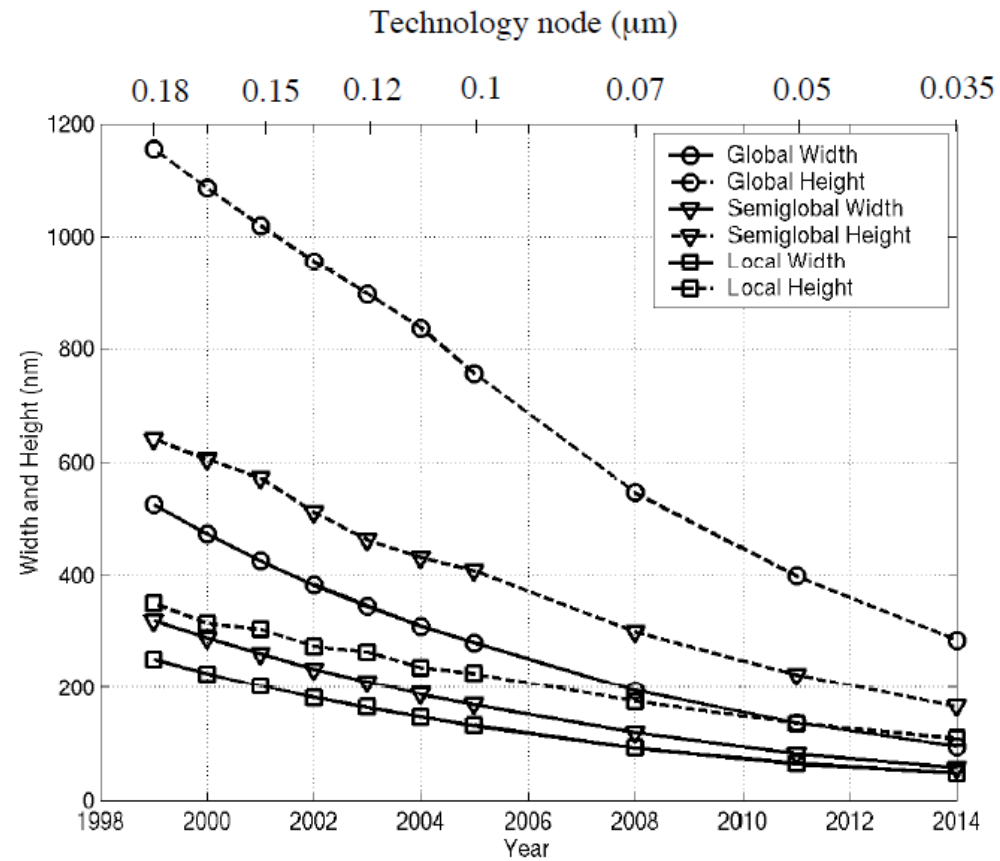
$$R = \rho \frac{L}{WH}$$

$$C_{ox} = K_{ox} \epsilon_0 \frac{WL}{X_{ox}}$$

$$C_I = K_{ox} \epsilon_0 \frac{HL}{L_S}$$

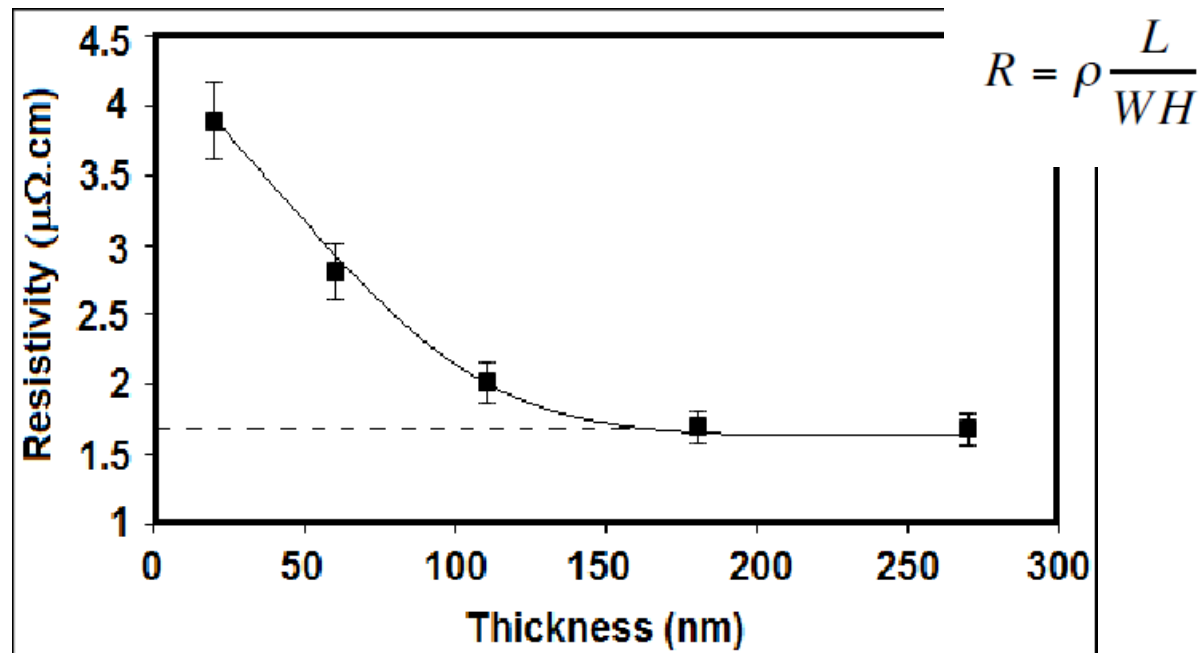
Ref: Prof. Saraswat, EE 311 Notes on interconnects, Stanford University

Interconnect scaling



Ref: Prof. Saraswat, EE 311 Notes on interconnects, Stanford University

Interconnect scaling

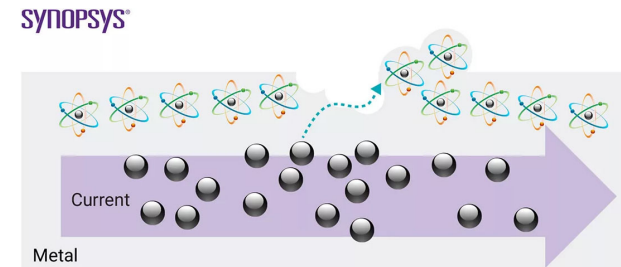


https://www.researchgate.net/figure/Variation-of-resistivity-for-copper-thin-films-as-function-of-film-thickness-at-room_fig6_266616142

Interconnect scaling

Electro-migration

- Electro-migration is the gradual displacement of metal atoms in an interconnect. It occurs when the current density is high enough to cause the drift of metal atoms in the direction of the electron flow.
- Electromigration is the main reliability issue in modern integrated circuits, which can trigger system failure at some undefined future time.
- The phenomenon is particularly likely to affect the thin, tightly spaced interconnect lines of deep-submicron designs.

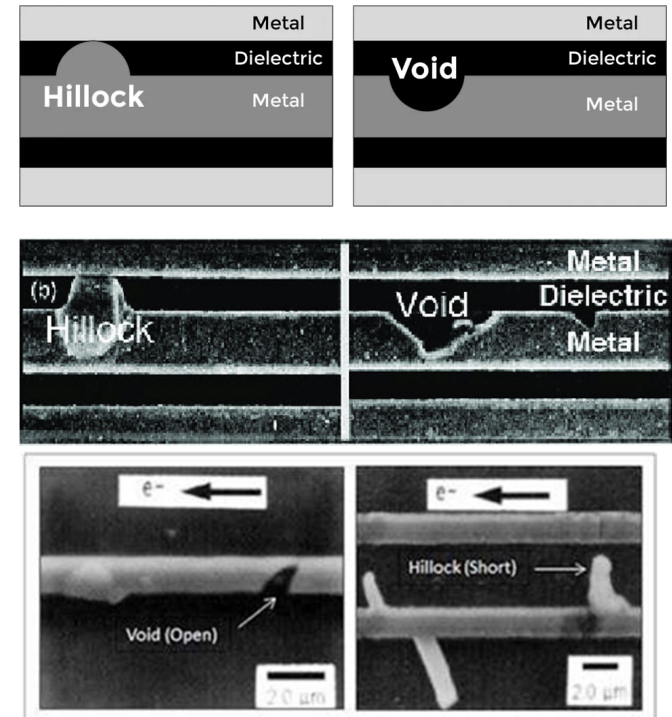


Interconnect scaling-Electromigration

- This will create both 'vacancies' and 'deposits'. The vacancies can grow and eventually break circuit connections resulting in open-circuits, while the deposits can grow and eventually close circuit connections resulting in short-circuit.
- Images of **a void and a hillock** formed in semiconductor interconnect through electromigration of metal atoms away from some areas and toward others

Where its Occurrence?

1. Interconnection of the metal
2. More Metal Bending
3. Thin metal rout at long distance
4. Scaling in interconnection



<https://www.ansys.com/blog/what-is-electromigration>

[Vittal T, Synopsis India](#)

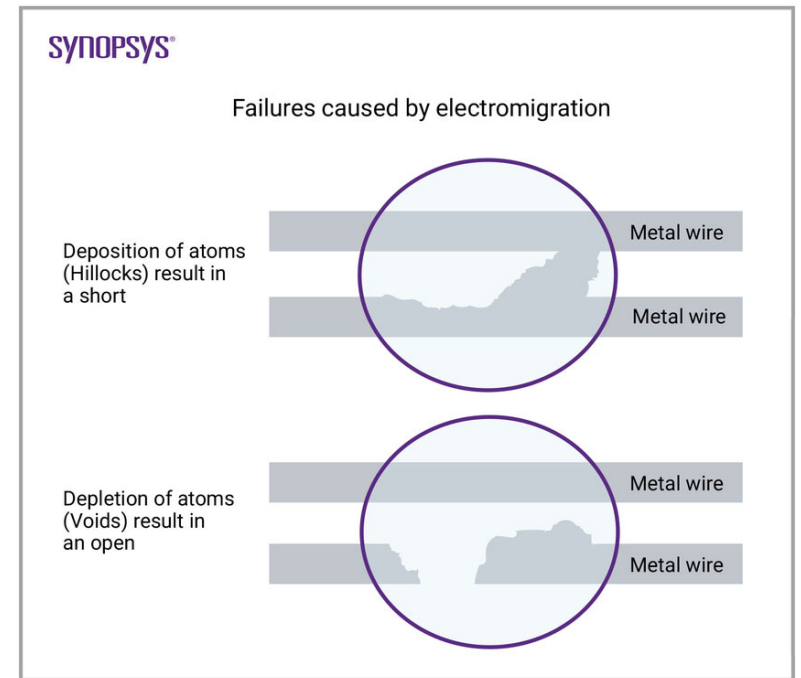
Interconnect scaling-Electromigration

Damage to the metallization lines is caused by *divergences* in atomic flux. When the amounts of matter leaving and entering a given volume are unequal, the associated accumulation or loss of material results in damage.

This results in two types of inequalities:

1. Depletion of atoms (Voids): Slow reduction of connectivity; Interconnect failure

2. Deposition of atoms (Hillocks): Shorts



Interconnect scaling-Electromigration

Mitigation Strategies:

To address electromigration issues, several strategies are employed:

- **Material Engineering:** Using barrier layers or different alloy compositions can help mitigate EM effects.
- **Temperature Management:** Effective thermal management techniques can help lower the operating temperature, thus reducing the rate of electromigration.
- **Testing and Modeling:** Advanced modeling and simulation techniques can help predict and analyze electromigration effects in designs, allowing engineers to optimize layouts before fabrication.
- **Redundant Paths:** Designing circuits with redundant interconnect paths can provide alternative routes for current, reducing the impact of potential voids.

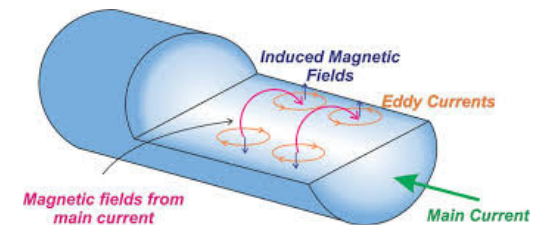
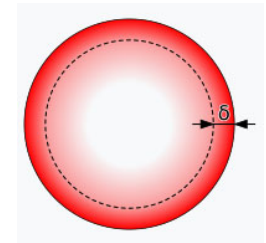
Interconnect scaling

Skin depth:

- In electromagnetism, **skin effect** is the tendency of an alternating current (AC) to become distributed within a conductor such that the current density is largest near the surface of the conductor and decreases exponentially with greater depths in the conductor.
- It is caused by opposing eddy currents induced by the changing magnetic field resulting from the alternating current. The electric current flows mainly at the *skin* of the conductor, between the outer surface and a level called the **skin depth**.

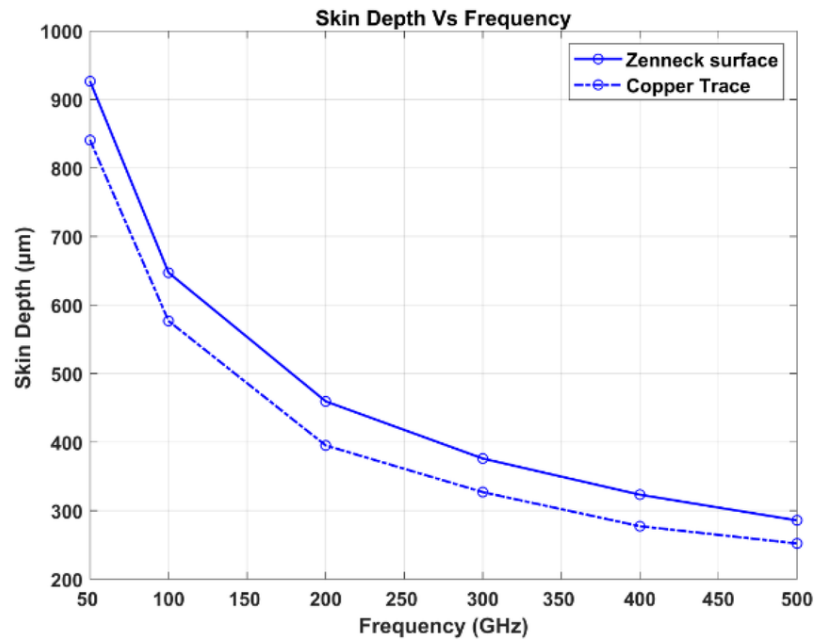
https://en.wikipedia.org/wiki/Skin_effect

<https://www.allaboutcircuits.com/technical-articles/understanding-the-skin-effect-cylindrical-and-rectangular-conductors-eddy-currents-and-current-crowding/>



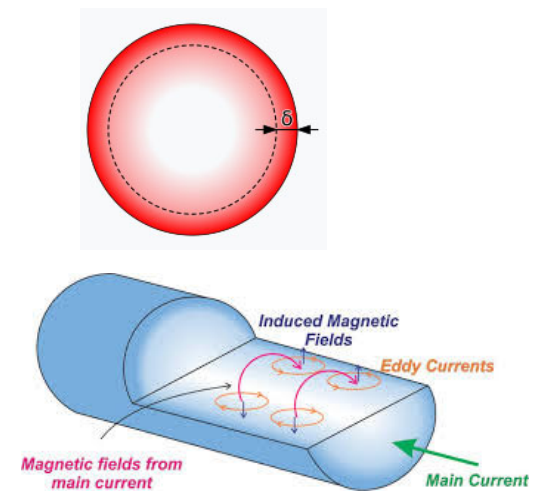
Interconnect scaling

Skin depth:



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https://en.wikipedia.org/wiki/Skin_effect



Interconnect scaling- Stress Migration

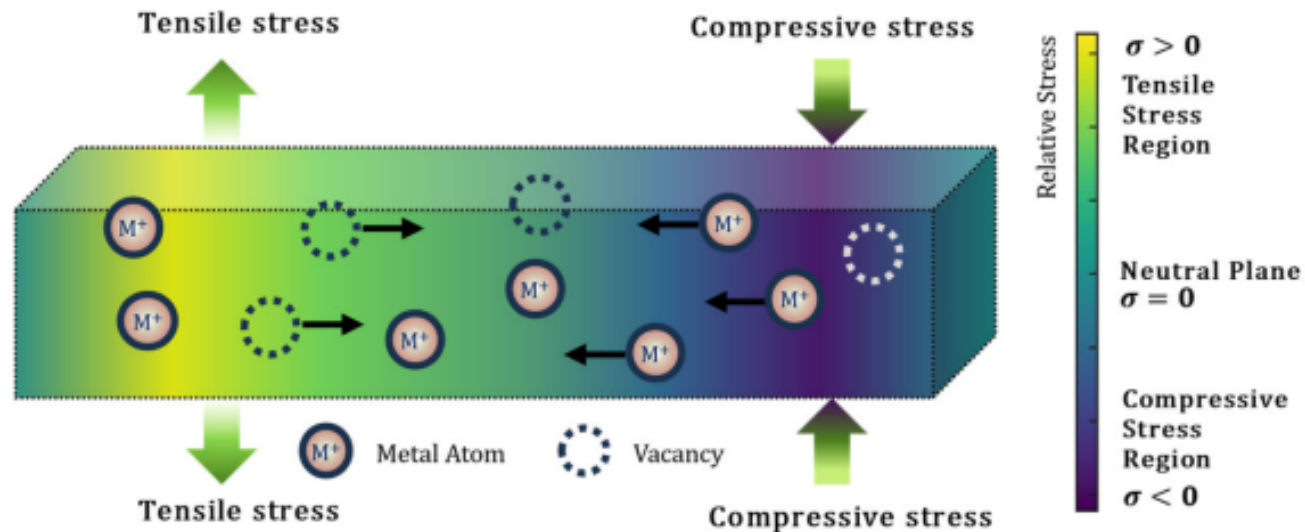
- **Stress Migration** is a reliability failure mechanism observed primarily in **metal interconnects** of CMOS integrated circuits. It occurs due to **mechanical stress gradients** within the metal lines, even *without any electric current* (unlike electromigration).
- Thermal-mechanical stresses cause vacancies to accumulate and form voids in metal lines and vias, potentially leading to circuit failure.
- These stresses arise from the difference in thermal expansion coefficients (CTE) between the [metal](#) and [dielectric materials](#).
- Stress migration is driven by stress gradients, leading to atomic mass transport and void formation, especially at locations with concentrated tensile stress, such as at via corners.

Interconnect scaling- Stress Migration

Physical Origin

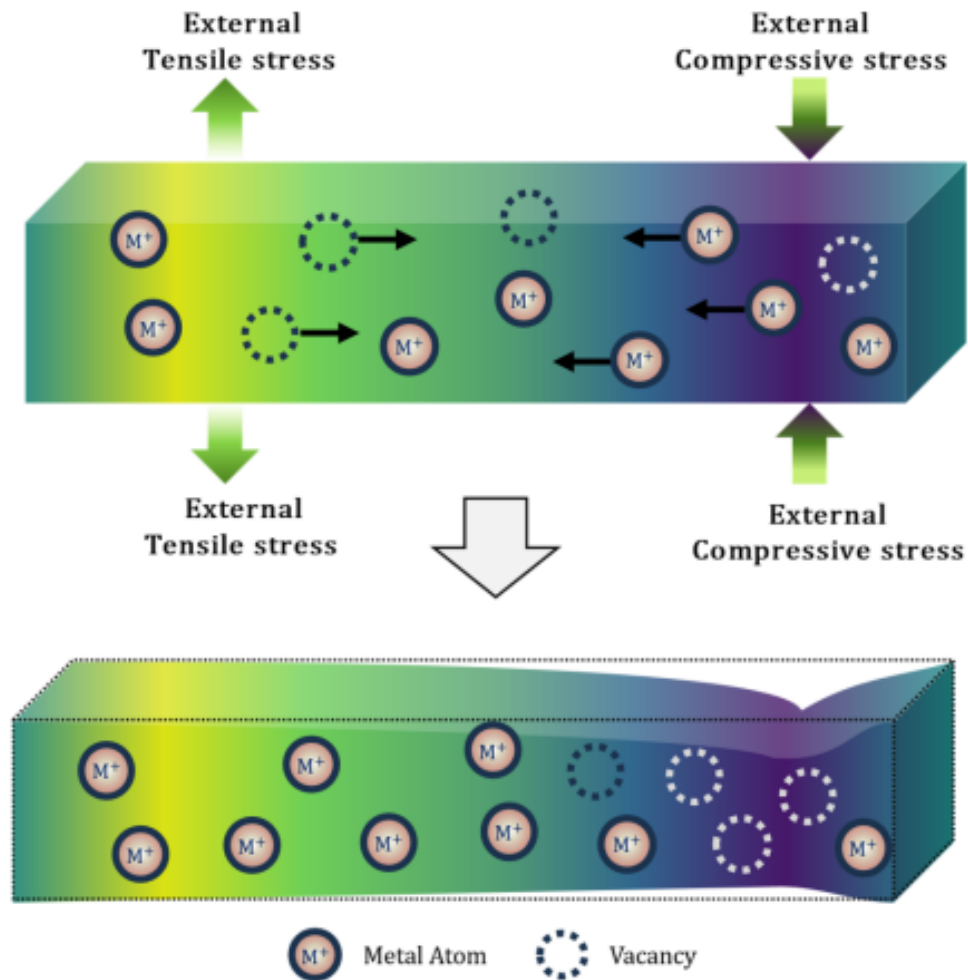
- During **deposition and cooling** of metal interconnects (like aluminum or copper), **thermal expansion mismatch** occurs between the metal and surrounding dielectric (e.g., SiO₂ or low-k material).
- As temperature decreases, **compressive stress** builds up in the metal.
- At elevated operating temperatures or during thermal cycling, **stress relaxation** takes place by atomic diffusion — atoms migrate from regions of **compressive stress** to **tensile stress** regions.
- This leads to **void formation** (depletion of atoms) or **hillock formation** (accumulation of atoms).

Interconnect scaling- Stress Migration



Schematic illustration of SM, depicting the movement of atoms driven by a stress gradient within the material.

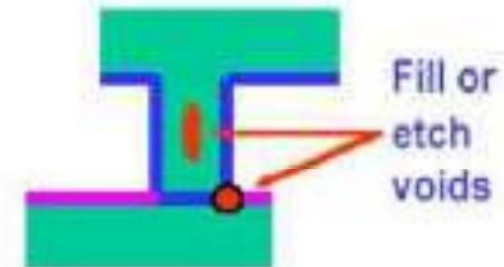
Interconnect scaling- Stress Migration



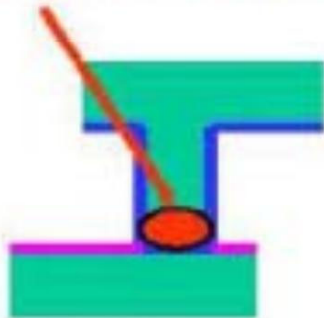
<https://doi.org/10.3390/electronics14153151>

Interconnect scaling- Stress Migration

(1) Void migration after 400°C PECVD cycle



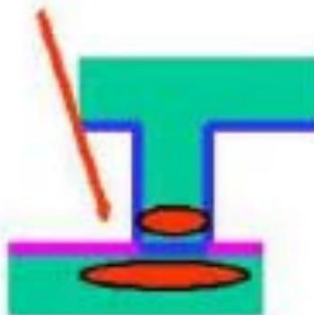
Voids migrate to bottom at 400°C



(2) Stress relaxation at 150°C - 200°C



Relaxation of stress = M1 or M2 voids



Interconnect scaling- Stress Migration

Aspect	Description
Driving force	Stress gradient ($\partial\sigma/\partial x$) — <i>not</i> current density
Dominant diffusion path	Grain boundary or interface diffusion
Temperature dependence	Exponential (Arrhenius) via diffusion coefficient
Typical locations	Near vias, line ends, or metal–dielectric interfaces

Interconnect scaling- Stress Migration

Difference from Electromigration (EM)

Feature	Stress Migration	Electromigration
Driving force	Stress gradient	Electron wind force
Current	Not required	Required
Direction	From compressive → tensile	Along electron flow
Temperature sensitivity	Moderate	High
Failure mode	Void or hillock	Open or short circuit

Interconnect scaling- Stress Migration

Mitigation Techniques

- Use of **Cu interconnects** with proper barrier layers (Ta/TaN).
- **Annealing** to reduce residual stress.
- **Capping layers** (e.g., SiN, CoWP) to control diffusion.
- **Optimized layout** (shorter line lengths, redundant vias).
- **Use of low-k dielectrics** to reduce thermal mismatch.

Interconnect scaling

Requirements of the interconnection materials

- Low resistivity of conductors
- Low capacitance => low dielectric constant
 - Low RC delay
 - Low cross talk
 - Low power dissipation (CV^2f loss)

Ref: Prof. Saraswat, EE 311 Notes on interconnects, Stanford University

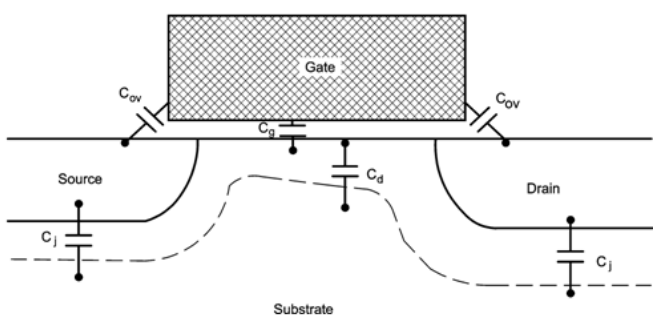
Interconnect scaling

- Resistance to electromigration
- Ease of deposition of thin films of the material
- Ability to withstand the chemicals and high temperatures required in the fabrication process
- Good adhesion to other layers - low physical stress
 - Stability of electrical contacts to other layers
 - Ability to contact shallow junctions and provide low resistance
 - Ability to be defined into fine patterns

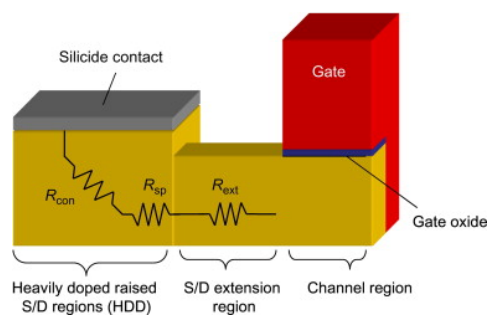
Ref: Prof. Saraswat, EE 311 Notes on interconnects, Stanford University

Interconnect scaling

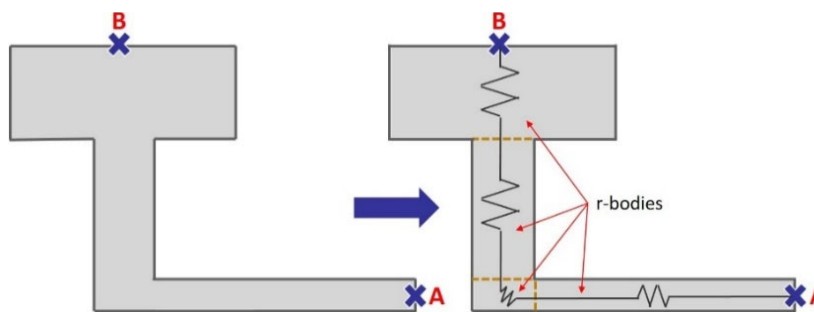
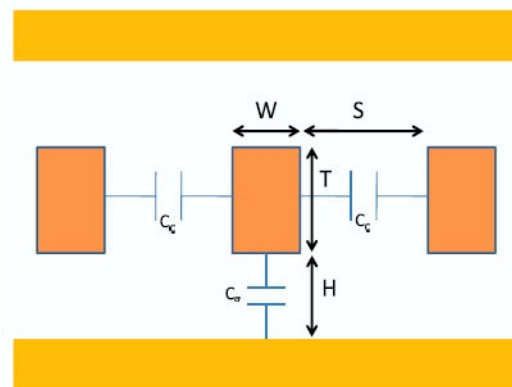
Parasitic Capacitances and Resistances



MOSFET parasitic capacitances



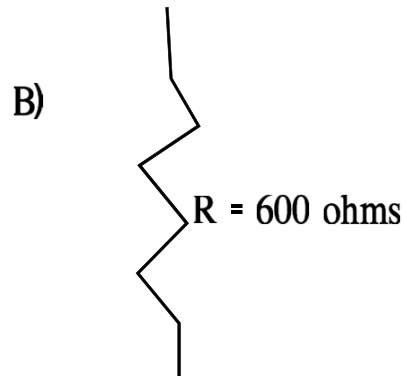
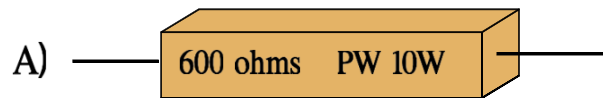
Interconnects



Resistors on ICs

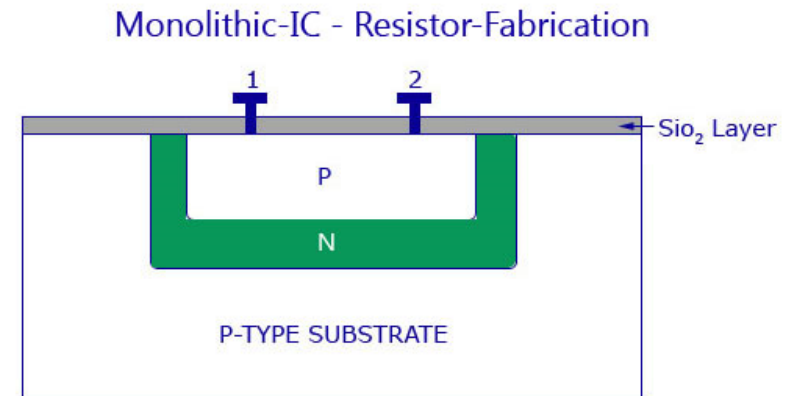
Resistors are fabricated on ICs using

- either the diffused resistor method- doping specific regions of the semiconductor material,
- or the thin-film resistor method, where a resistive thin film is deposited and patterned on the substrate



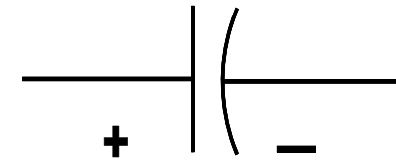
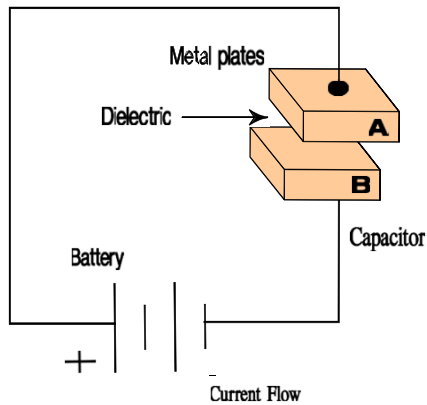
$$R = \frac{\rho L}{A}$$


- Key Theme:
- Resistance is measured in **ohms**



Capacitors on ICs

- Capacitance is measured in **farads** and contains two conductive plates and a dielectric between the two plates



Symbol for a capacitor

Capacitor in DRAM

